Gain-weighted well-counting in a congestion mean-field game

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Abstract

We prove, by validated numerics, that a stationary discounted congestion mean-field game on the torus admits equilibria whose population density has strictly more local maxima than the cost potential has wells. With $V(x) = A_1 \cos 2\pi x + A_2 \cos 4\pi x$ and $A_1 > 4A_2$—exactly one well—the certified equilibrium density at our main instance has exactly 2 strict local maxima (Theorem 1); a third-harmonic instance yields exactly 3 (Theorem 2). The mechanism is elementary and we state it as such: the equilibrium re-weights harmonics *the potential already contains* through the band-pass linear-response gain $c(\kappa) = \kappa/[(1 + \sigma\kappa)^2 + \gamma\kappa]$, so the density counts wells with gain-weighted amplitudes while the potential counts them flat. The second-versus-first harmonic crossover $\sigma^* = 1/(8\pi^2)$ is independent of the coupling γ—decided here as an exact-rational polynomial identity (the γ-coefficient of the crossover polynomial is identically zero), not by float agreement—and the limiting splitting window is the exact rational interval (1/16, 1/4) (third harmonic: (1/27, 1/3), the Chebyshev U_{k-1} law). Around the two theorems sit six further certified instances forming a bracket table: negatives below the amplitude threshold and above σ^*, replications, and a threshold pin—the splitting threshold at $\sigma = 0.002$, $\gamma = 0.01$ lies in $[0.13, 0.14]$ by certified counts, with the exact-rational linear-response prediction $r_c = 0.132724608698\ldots$ inside. All enclosures are radii-polynomial certificates at radii between 2.97×10^{-14} and 5.32×10^{-13}, with local uniqueness in the full sequence-space ball (even and odd blocks) and both positivity walls certified. Every number in this document is interpolated from those certificates.

1 Introduction

Under Lasry–Lions monotonicity [1], mean-field-game theory guarantees a unique equilibrium, and the intuition that travels with the theory is stronger than the theorem: the population's density is expected to mirror the cost—wells of the potential become peaks of the crowd, one for one. The known rigorous results reinforce that reading (in the Cesaroni–Cirant regime [2], density peaks sit where the potential's wells sit), and the natural working assumption is a *ceiling*: the equilibrium cannot carry more local maxima than the cost has wells.

This paper proves the ceiling false, by validated numerics, for a congestion system with a monotone coupling. At explicit parameters of a stationary discounted congestion mean-field

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game on the torus we certify exact equilibria whose density carries exactly two—and, at a third-harmonic instance, exactly three—strict local maxima over a potential with exactly one well. The mechanism is not exotic and we state it at its true altitude: the linearized response to the cost is a band-pass filter in harmonic frequency, so the equilibrium counts wells with *gain-weighted* amplitudes while the potential counts them flat, and between those two counts a single well splits the crowd. What the linear analysis cannot give—and the certificates do—is the nonlinear statement: these are exact equilibria of the full congestion system, with local uniqueness in the full sequence-space ball and both positivity walls certified. The uniqueness is the certificate's own, local to the enclosure ball; no uniqueness theorem for the congestion system is invoked.

Contributions. (i) The phenomenon theorem and its three-peak companion (Sections 4–5), by radii-polynomial enclosures whose certified balls fix the exact critical-point count of the exact solution. (ii) The crossover constant $\sigma^* = 1/(8\pi^2)$, independent of the coupling γ, decided as an exact-rational polynomial identity rather than observed numerically (Section 3). (iii) A bracket table of 6 further certified instances pinning the splitting threshold inside $[0.13, 0.14]$, with the exact-rational linear-response prediction landing inside the pin (Section 6). (iv) An exact solution count for the companion ergodic system—exactly 3 Galerkin solutions, box-bounded—together with function-space multiplicity: at least three distinct exact solutions per coupling in pairwise disjoint uniqueness balls (Section 7). Every number in this document is interpolated at build time from machine-written certificates, named where used; the build refuses to render if any certificate is missing, refused, or structurally moved (Section 8 and the reproduction note).

Honest counting. The finding is two theorems and one table. Rows of the bracket table are rows of a table, not further theorems; the companion results are stated at their own level—a truncation-level count, and a function-space lower bound.

2 The model and the exact instances

The stationary discounted congestion mean-field game on the unit torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with congestion exponent $a = 1/2$, is the system

$$u - \sigma u'' + \tfrac{1}{2}\,\frac{(u')^2}{\sqrt{m}} = V(x) + \gamma m, \tag{1}$$

$$m - \sigma m'' - \left(\sqrt{m}\,u'\right)' = 1, \qquad m > 0, \tag{2}$$

with the congestion Hamiltonian $H(p, m) = \frac{1}{2}|p|^2 m^{-1/2}$, viscosity $\sigma > 0$ and the coupling γm with $\gamma > 0$, which is increasing in m: the crowd pays for density. The discount enters both equations as the zero-order terms at unit rate, and the Fokker–Planck equation (2) carries the uniform unit source of a population renewed at that rate. The potential is the trigonometric polynomial

$$V(x) = A_1 \cos 2\pi x + A_2 \cos 4\pi x + A_3 \cos 6\pi x. \tag{3}$$

The system is augmented with the branch variable $w = m^{-1/2}$ and the reciprocal constraint $m\,w^2 = 1$, so that the congestion terms become polynomial in the unknowns, and solved in even cosine Fourier space—u, m and w are even—with the derivative field $p = u'$ carried as an odd (sine) sequence. The unknown is the tuple (a_0, p, m, w): the mean of u and the coefficient sequences of p, m and w.

Coefficient honesty. Each theorem is a statement about specific binary64 doubles, not about the decimal strings that name them. The main instance (T1) and the three-peak instance (T6) share σ, γ, A_1, the truncation N and the weight ν:

parameter	decimal name	exact binary64	instance
σ	0.002	`0x1.0624dd2f1a9fcp-9`	T1, T6
γ	0.01	`0x1.47ae147ae147bp-7`	T1, T6
A_1	0.003	`0x1.89374bc6a7efap-9`	T1, T6
A_2	0.0006000000000000001	`0x1.3a92a30553262p-11`	T1 ($A_3 = 0$)
A_3	0.00075	`0x1.89374bc6a7efap-11`	T6 ($A_2 = 0$)

with $N = 96$ cosine modes, $\nu = 1.02$, and 1,536 grid points in the positivity certificates. The value of A_2 is the binary64 product 0.003×0.2 as the origin computation evaluated it, which lies 1 unit in the last place above the double nearest 0.0006; the theorem is about that double. At T1, $A_2/A_1 = 0.2$, so $A_1 > 4A_2$ and V has exactly one well; at T6, $A_3/A_1 = 0.25 < 1/3$ and again V has exactly one well (Proposition 3). Both well counts are also certified directly, by the same instrument that counts the density's peaks (Sections 4–5).

3 Linear response: the band-pass gain and the exact crossover

At first order in the potential amplitudes the congestion term drops out and each harmonic responds independently. Linearize (1)–(2) about the flat state $(u, m) = (\gamma, 1)$ of $V \equiv 0$: writing $u = \gamma + \hat{u}$, $m = 1 + \hat{m}$ and keeping first-order terms,

$$\hat{u} - \sigma\hat{u}'' = V + \gamma\hat{m}, \qquad \hat{m} - \sigma\hat{m}'' = \hat{u}''.$$

For the k-th cosine mode, with $\kappa_k = (2\pi k)^2$, this reads $(1+\sigma\kappa_k)\hat{u}_k = A_k + \gamma\hat{m}_k$ and $(1+\sigma\kappa_k)\hat{m}_k = -\kappa_k\hat{u}_k$, whence $\hat{m}_k = -c(\kappa_k)A_k$ with

$$c(\kappa) = \frac{\kappa}{(1+\sigma\kappa)^2 + \gamma\kappa}. \tag{4}$$

To first order the density is $m \approx 1 - \sum_k c(\kappa_k)A_k \cos 2\pi kx$: the crowd's harmonics are the potential's, each multiplied by the gain $c(\kappa_k)$ and reflected. Three facts about (4) are decided in exact rationals (`certs/terra-sigmastar.json`; standard-library fractions, no floating point in any verdict).

Proposition 1 (band-pass)**.** *Let $D(\kappa) = (1+\sigma\kappa)^2 + \gamma\kappa$ be the denominator of* (4)*. As polynomials in $\mathbb{Q}[\sigma, \kappa, \gamma]$,*

$$D - \kappa\,\partial_\kappa D = (1+\sigma\kappa)(1-\sigma\kappa) = -\sigma^2\kappa^2 + 1.$$

The γ-term cancels identically, so $\operatorname{sign} c'(\kappa) = \operatorname{sign}(1-\sigma\kappa)$ for every γ: the gain peaks at $\kappa = 1/\sigma$.

Proposition 2 (γ-free crossover)**.** *With $s = \sigma\kappa_1$ and $g = \gamma\kappa_1$, the k-th-versus-first crossover polynomial*

$$N_k(s, g) = k^2\big[(1+s)^2 + g\big] - \big[(1+k^2 s)^2 + k^2 g\big]$$

factors as $(k^2-1)(1-k^2s^2)$: its g-coefficient is the zero polynomial and its positive root is $s = 1/k$ exactly, with $N_k > 0$ below the root and $N_k < 0$ above it (decided for $k = 2, \ldots, 12$). Hence $c(\kappa_k) = c(\kappa_1)$ exactly at $\sigma = 1/(4\pi^2 k)$; at $k = 2$,

$$\sigma^* = \frac{1}{8\pi^2} = 0.012665147955292221430484932901\ldots \tag{5}$$

The decimal digits in (5) are read from an exact rational bracket of σ^* of width below 1.27×10^{-44}, obtained from a rational bracket of π (Machin's formula with alternating-series bounds) and outward division; the certificate carries the rational endpoints, and the digits shown are those the two endpoints share. The γ-independence is an exact cancellation, not numerical smallness. The crossover polynomials as the certificate prints them after the cancellation of g are

$$N_2 = 3 - 12\,s^2, \quad N_3 = 8 - 72\,s^2, \quad N_4 = 15 - 240\,s^2, \quad N_5 = 24 - 600\,s^2,$$
$$N_6 = 35 - 1260\,s^2, \quad N_7 = 48 - 2352\,s^2, \quad N_8 = 63 - 4032\,s^2, \quad N_9 = 80 - 6480\,s^2,$$
$$N_{10} = 99 - 9900\,s^2, \quad N_{11} = 120 - 14520\,s^2, \quad N_{12} = 143 - 20592\,s^2.$$

Proposition 3 (the splitting windows)**.** *Since* $\sin 2\pi kx = \sin 2\pi x\ U_{k-1}(\cos 2\pi x)$*, with* U_{k-1} *the Chebyshev polynomial of the second kind, the two-harmonic potential* $V = A_1 \cos 2\pi x + A_k \cos 2\pi kx$ *with* $r = A_k/A_1 > 0$ *has*

$$V'(x) = -2\pi A_1 \sin 2\pi x \left[1 + k\,r\,U_{k-1}(\cos 2\pi x)\right],$$

so V *acquires interior critical points beyond* $x = 0, \frac{1}{2}$ *if and only if* $k\,r\,|\min U_{k-1}| > 1$*. The flat thresholds are exact rationals read from the ranges of* $U_1 = 2c$ *and* $U_2 = 4c^2 - 1$ *on* $(-1, 1)$*:* $r = 1/4$ *at* $k = 2$ *and* $r = 1/3$ *at* $k = 3$*. To first order the equilibrium density obeys the same criterion with the gain-weighted ratio* $r_{\text{eff}} = r\,c(\kappa_k)/c(\kappa_1)$*, and* $c(\kappa_k)/c(\kappa_1) \to k^2$ *as* $(\sigma, \gamma) \to 0$ *(the limits* 4 *and* 9 *are exact evaluations), so the limiting splitting windows—where the density splits while the potential does not—are exactly*

$$\big(1/16,\ 1/4\big) \text{ at } k = 2, \qquad \big(1/27,\ 1/3\big) \text{ at } k = 3.$$

One sentence, stated honestly: between the gain-weighted count and the flat count, the crowd splits at the bottom of a single well. The mechanism—a non-flat linear gain changing the critical-point count of a two-harmonic signal—is elementary, and Proposition 3 is its complete statement; stating it as elementary isolates the actual contribution, which is the mean-field-game setting, the exact γ-free constant, and the nonlinear certificates below. Linear response predicts; only the certificates prove.

4 Theorem 1: two peaks over one well

Write ℓ^1_ν for the weighted sequence space with norm $\|f\|_\nu = |f_0| + 2\sum_{k\geq 1} |f_k|\nu^k$ on even (cosine) sequences, and the same weights on odd (sine) sequences; the augmented unknown (a_0, p, m, w) lives in a product of such spaces, and "the ball" below is the ball of radius r about the stored candidate in that augmented space.

Theorem 1 (two peaks over one well)**.** *At the instance T1 of Section 2 (*$A_1 > 4A_2$*: V has exactly one well) there is an exact equilibrium* (u, m, w) *of the full congestion system* (1)–(2) *within* ℓ^1_ν *distance* $r = 2.520 \times 10^{-13}$ *of the stored candidate, locally unique in the full sequence-space ball—even (cosine) and odd (sine) blocks alike—with* $m \geq 0.8943$ *and* $w \geq 0.9742$ *over the whole ball. Every density in that ball has exactly 2 strict local maxima and 2 strict local minima on the torus, while V has exactly 1 well. (Records:* `certs/terra-recert-t1.json`*,* `certs/terra-peakcount-t1.json`*.)*

The enclosure. The radii-polynomial (Newton–Kantorovich) argument [6] fixes an approximate inverse A of the Jacobian $DF(\bar{x})$ at the candidate $\bar{x}$ and bounds, in outward-rounded interval arithmetic, the defect $Y_0 \geq \|A\,F(\bar{x})\|$, the contraction constant $Z_1 \geq \|I - A\,DF(\bar{x})\|$ (taken over the even and the odd block) and a second-order constant Z_2; a radius r with

$$p(r) = \tfrac{1}{2} Z_2\, r^2 - (1 - Z_1)\, r + Y_0 < 0 \tag{6}$$

yields a unique zero of F in the ball of radius r, and the same zero is the unique one in the ball of any radius in $(r_{\min}, r_{\max})$, the roots of p. The T1 numbers are

$$Y_0 = 2.473 \times 10^{-14}, \qquad Z_1 = 0.8970 \text{ (even and odd blocks alike)}, \qquad Z_2 = 1131.8244,$$

with admissible radii $(r_{\min}, r_{\max}) \supseteq (2.400 \times 10^{-13},\ 1.820 \times 10^{-4})$; the certificate takes r one factor-1.05 step above $r_{\min}$ and verifies $p(r) < 0$ in interval arithmetic with margin 4.690×10^{-6}. The tail of the sequence space is controlled through the Banach-algebra property of ℓ^1_ν, so the enclosed object solves the system, not an N-mode approximation; and the odd block is certified because that is where a symmetry-breaking solution would live—evenness of the equilibrium is a corollary of the full-ball uniqueness, not an assumption. Positivity of m and of the branch selector w is certified over the whole ball by an interval evaluation of the candidate on 1,536 grid points with a modulus-of-continuity slack, less the radius: $m \geq 0.8943$ and $w \geq 0.9742$ everywhere in the ball, which pins the physical branch $w = +m^{-1/2}$. The floors are stated rounded down, the radii and contraction constants rounded up, here and throughout.

The count. The peak count derives only from certified region signs. For an even function f on $\mathbb{T}$ it suffices to work on $[0, \frac{1}{2}]$, where $f'(0) = f'(\frac{1}{2}) = 0$. A float scan proposes the interior critical points and the inflection gaps; the machine then builds an alternating chain of regions—a curvature region on which the sign of f'' is certified, a slope region on which the sign of f' is certified, and so on—covering $[0, \frac{1}{2}]$ contiguously, and asserts on the certified signs that each slope sign equals the curvature sign before it and negates the one after it. That forces exactly one zero of f' in each curvature region and none elsewhere, so the critical points on $[0, \frac{1}{2}]$ are exactly one per curvature region, with kinds given by the certified curvature signs; torus counts follow by evenness (interior points mirror, the endpoints do not). Every sign is certified over the whole ball: the derivative pads $\sup_{\text{ball}} |\delta m'| \leq 2.941 \times 10^{-11}$ and $\sup_{\text{ball}} |\delta m''| \leq 1.374 \times 10^{-8}$, which follow from $\|\delta m\|_\nu \leq r$ through $\sup |\delta m^{(j)}| \leq r\,(2\pi)^j \max_k k^j \nu^{-k}$, are folded into every cell bound, every coefficient product is enclosed outward, and the rounding chain of the trigonometric arguments is bounded. Each region is certified on an adaptive grid of 256 cells. The T1 chain for the density is:

region	interval	certified sign	margin
f''	$[0.0000,\ 0.1065]$	$+$	3.258×10^{0}
f'	$[0.1065,\ 0.2154]$	$+$	6.100×10^{-1}
f''	$[0.2154,\ 0.4058]$	$-$	7.105×10^{-1}
f'	$[0.4058,\ 0.4469]$	$-$	6.724×10^{-2}
f''	$[0.4469,\ 0.5000]$	$+$	7.482×10^{-1}

Three curvature regions with signs $+, -, +$ give, on $[0, \frac{1}{2}]$, a minimum at 0, a maximum inside the certified region $[0.2153,\ 0.4059]$ (the float proposal is $x \approx 0.363$) and a minimum at $\frac{1}{2}$; on the torus, maxima at $x \approx 0.363, 0.637$ and minima at $x = 0, 1/2$: exactly 2 strict maxima and 2 strict minima. The smallest certified margin, 6.724×10^{-2}, is more than 6 orders of magnitude above the pads. The same chain for V,

region	interval	certified sign	margin
f''	$[0.0000,\ 0.1217]$	$-$	8.938×10^{-2}
f'	$[0.1217,\ 0.2717]$	$-$	1.661×10^{-2}
f''	$[0.2717,\ 0.5000]$	$+$	2.282×10^{-2}

certifies that V has exactly one well, at $x = \frac{1}{2}$, so the theorem's "one well" is a certified count too, not only the consequence of $A_1 > 4A_2$. The region breakpoints are the proposer's floats, shown to four decimals; the record holds them exactly, and the signs are certified on the exact intervals.

5 Theorem 2: three peaks over one well

Theorem 2 (three peaks over one well)**.** *At the instance T6 of Section 2 (third harmonic, $A_3/A_1 = 0.25$, $A_2 = 0$: V has exactly one well) the analogous statement holds with radius $r = 2.732 \times 10^{-13}$, $m \geq 0.8823$ and $w \geq 0.9461$ over the whole ball, and every density in the ball has exactly 3 strict local maxima and 3 strict local minima on the torus, while V has exactly 1 well. (Records:* `certs/terra-recert-t6.json`*,* `certs/terra-peakcount-t6.json`*.)*

The enclosure numbers are $Y_0 = 2.481 \times 10^{-14}$, $Z_1 = 0.9047$ (both blocks), $Z_2 = 1150.4173$, admissible radii $\supseteq (2.602 \times 10^{-13},\ 1.657 \times 10^{-4})$ and closure margin 3.952×10^{-6}. The chain has 7 regions:

region	interval	certified sign	margin
f''	$[0.0000,\ 0.0642]$	$+$	7.809×10^{0}
f'	$[0.0642,\ 0.1240]$	$+$	8.893×10^{-1}
f''	$[0.1240,\ 0.2341]$	$-$	3.773×10^{0}
f'	$[0.2341,\ 0.2641]$	$-$	2.662×10^{-1}
f''	$[0.2641,\ 0.3750]$	$+$	3.773×10^{0}
f'	$[0.3750,\ 0.4352]$	$+$	8.807×10^{-1}
f''	$[0.4352,\ 0.5000]$	$-$	7.667×10^{0}

Four curvature regions with signs $+, -, +, -$ give a minimum at 0, then a maximum, a minimum and a maximum at $\frac{1}{2}$; the interior maximum lies in the certified region $[0.1239,\ 0.2341]$, and the float proposals for the two interior critical points are $x \approx 0.199, 0.299$. On the torus, maxima at $x \approx 0.199, 1/2, 0.801$ and minima at $x \approx 0, 0.299, 0.701$: exactly 3 strict maxima and 3 strict minima, with smallest certified margin 2.662×10^{-1} against pads 3.188×10^{-11} and 1.489×10^{-8}—7 orders of magnitude. The potential's own chain certifies one well (smallest margin 4.445×10^{-3}): $A_3/A_1 = 0.25$ lies inside the third-harmonic window $(1/27,\ 1/3)$ and below the flat threshold $1/3$, which is the splitting the crowd then performs.

6 The bracket table: the certified instances under the theorems

Honest counting: the finding is the two theorems above plus this table—rows of a table are rows of a table, not further theorems. Each row is an enclosure with full-ball local uniqueness (even and odd blocks) and a certified critical-point count, built by the same two instruments; every bound is shown (Table 1).

instance	σ	$r_k = A_k/A_1$	N	ν	radius r	Z_1	closure	$\min m$	peaks / wells
T1	0.002	$r_2 = 0.2$	96	1.02	2.520×10^{-13}	0.8970	4.690×10^{-6}	0.8943	2 / 1
T2	0.002	$r_2 = 0.12$	96	1.02	2.247×10^{-13}	0.8889	5.541×10^{-6}	0.9058	1 / 1
T3	0.02	$r_2 = 0.2$	64	1.05	2.967×10^{-14}	0.5235	4.985×10^{-4}	0.9620	1 / 1
T4	0.002	$r_2 = 0.15$	96	1.02	2.368×10^{-13}	0.8920	5.213×10^{-6}	0.9015	2 / 1
T5	0.001	$r_2 = 0.15$	176	1.015	5.314×10^{-13}	0.9260	2.106×10^{-6}	0.9649	2 / 1
T6	0.002	$r_3 = 0.25$	96	1.02	2.732×10^{-13}	0.9047	3.952×10^{-6}	0.8823	3 / 1
T7	0.002	$r_2 = 0.13$	96	1.02	2.301×10^{-13}	0.8899	5.430×10^{-6}	0.9044	1 / 1
T8	0.002	$r_2 = 0.14$	96	1.02	2.324×10^{-13}	0.8909	5.321×10^{-6}	0.9029	2 / 1

Table 1: The bracket table (`certs/terra-bracket-table.json`; $\gamma = 0.01$ throughout). The ratio column is $r_2 = A_2/A_1$ with $A_3 = 0$, or $r_3 = A_3/A_1$ with $A_2 = 0$; "closure" is the margin by which $p(r) < 0$ in (6). Radii and Z_1 are rounded up, margins and density floors rounded down. T1 and T6 are the theorems; T2 ($r = 0.12$) and T3 ($\sigma = 0.02 > \sigma^*$) are one-peak negatives exactly where the linear response predicts none; T4 and T5 ($\sigma = 0.001$, $N = 176$) replicate the split at $r = 0.15$; T7 and T8 pin the threshold. The tightest ball is T3's, at 2.97×10^{-14}; the lowest certified density floor over all 8 balls is 0.8823.

The threshold pin. T7 ($r = 0.13$, one peak) and T8 ($r = 0.14$, two peaks) pin the splitting threshold at $\sigma = 0.002$, $\gamma = 0.01$ inside $[0.13, 0.14]$ by certified counts alone: every crossing of the threshold lies between them. The linear-response prediction for the threshold—the ratio r at which $r_{\text{eff}} = r\, c(\kappa_2)/c(\kappa_1)$ reaches the flat threshold $1/4$—is

$$r_c = \frac{(1+4s)^2 + 4g}{16\big[(1+s)^2 + g\big]} = 0.132724608698\ldots, \qquad s = 4\pi^2\sigma,\ g = 4\pi^2\gamma, \tag{7}$$

decided as an exact rational bracket (the rational bracket of π again; the record holds the endpoints and the digits shown are those they share), and it lands inside the pin. The prediction is linear response; the pin is carried by the certified counts alone. T3 sits past the crossover ($\sigma = 0.02 > \sigma^*$), where Proposition 2 makes the second harmonic's gain smaller than the first's, and its single peak is the negative the proposition predicts; T2 sits below r_c and T4 above it, on the predicted sides.

7 Companion results: an exact count, and certified multiplicity

The companion system is the ergodic quadratic mean-field game on the torus with no potential,

$$-\sigma u'' + \tfrac{1}{2}(u')^2 + \rho = c\, m, \qquad -\sigma m'' - (m\, u')' = 0, \qquad \int m = 1,\ \int u = 0,\ m > 0, \tag{8}$$

with the ergodic constant ρ among the unknowns and a coupling c that is crowd-averse (monotone) for $c \geq 0$ and herding for $c < 0$. The symmetry-broken branch bifurcates from the constant state $(u, m) = (0, 1)$ at the pitchfork $c^* = -\sigma^2(2\pi)^2 = -\pi^2 \approx -9.8696$ at $\sigma = 1/2$.

An exact count at truncation level. At $\sigma = 1/2$ and $c = -12$, a Krawczyk exhaustion census proves that the N-mode even Galerkin truncation of (8)—unknowns $\rho, a_1, \ldots, a_N, b_1, \ldots, b_N$, with 2π carried as an enclosure—has exactly 3 solutions in an explicit printed box for $N = 2, 3, 4, 5$: in an adaptive partition of the box every subbox is eliminated by an interval-residual or a Krawczyk exclusion, and each surviving box is isolated by the Moore–Krawczyk condition

$K(X) \subset \operatorname{int} X$, existence and uniqueness per box [7]; the three surviving boxes are matched one-to-one to the constant solution, the symmetry-broken branch and its half-shift mirror.

N	dimension	boxes processed	by residual	by Krawczyk	seconds
2	5	2,077	824	169	0.1
3	7	21,642	9,742	945	0.8
4	9	335,862	157,608	9,399	17.7
5	11	6,954,073	3,322,545	143,471	586.7

The claim is stated box-bounded and at truncation level; the PDE-level count is an open problem, not an implication. (Records: `certs/mfg-cap-census-N{2..5}-c-12.json`.)

Function-space multiplicity. At each of six couplings $c \in$ {-11,-12,-14,-16,-20,-24}—all below the pitchfork c^*, far from the monotone half-line—the constant solution, the symmetry-broken branch and its half-shift mirror are enclosed in pairwise disjoint ℓ^1_ν uniqueness balls ($\nu = 1.02$) with certified positive density: at least three distinct exact solutions of (8) at every listed coupling. The half-shift $x \mapsto x + \frac{1}{2}$ maps solutions to solutions when $V \equiv 0$, and the mirror candidate is that image of the branch, so once the two sit in disjoint balls the symmetry is proved to produce a different solution, not a relabeling. At $c = -9.5$, above the pitchfork, the branch continuation collapses onto the constant and no claim is made—the recorded boundary.

c	N	r_{const}	r_{branch}	$\max Z_1$	$\min m$	gap
-11	16	7.05×10^{-15}	4.73×10^{-13}	0.273	2.973×10^{-1}	6.70
-12	16	7.22×10^{-15}	1.63×10^{-11}	0.321	1.629×10^{-1}	10.78
-14	16	7.23×10^{-15}	5.90×10^{-9}	0.409	5.633×10^{-2}	16.76
-16	16	1.34×10^{-14}	2.14×10^{-7}	0.495	1.903×10^{-2}	21.43
-20	24	1.51×10^{-14}	9.27×10^{-9}	0.441	3.282×10^{-3}	30.51
-24	32	1.46×10^{-14}	1.09×10^{-9}	0.415	4.446×10^{-4}	39.85

Here r_{const} is the radius of the constant solution's ball, r_{branch} that of the branch and of its mirror (equal to the printed digits), $\max Z_1$ the largest contraction constant of the three, $\min m$ the certified density floor over the branch and mirror balls, and the gap the certified ℓ^1_ν distance between the nearest two balls less the sum of their radii. The radii grow and the density digs toward vacuum as the coupling deepens, down to $\min m \geq 4.446 \times 10^{-4}$ at $c = -24$. (Record: `certs/mfg-cap-multiplicity.json`.)

8 Method and verification posture

The enclosure instrument. The enclosures are radii-polynomial certificates on the augmented (a_0, p, m, w) system with the reciprocal constraint, with ℓ^1_ν Banach-algebra tail bounds and both blocks of the linearization certified. The verifier is the frozen standard-library Python verifier that ships, byte for byte, inside the companion page of the program's base instance, `reports/mfg-congest.html`; the file re-extracted from that page is pinned by its sha256, whose first digits are `7c5356222d62`. A living module extends it by only the second- and third-harmonic data terms of V and the construction of the approximate inverses by Gauss–Jordan elimination. The extension is battery-pinned to the frozen file: on the frozen file's own embedded instance the extended validation must reproduce the frozen one bit for bit. An approximate inverse is only ever a *candidate* for a contraction—every bound is recomputed from it rigorously, so a bad inverse fails $Z_1 < 1$ and cannot fake a proof.

Two implementations. The candidates were first enclosed by a JavaScript kernel at the origin bench; the certificates here are the re-certification by the Python lineage, with independently computed approximate inverses on each side. On T1 the two agree on the certified radius exactly (ratio 1 to the last digit) and on Z_1 and on the floor of w exactly; the density floors differ by 0.00017 (origin 0.8945, here 0.8943), because the positivity routine here subtracts a modulus-of-continuity slack the origin did not—ours is the lower, and it is the one stated. Float witnesses accompany the certificate and are labeled as such: on a pointwise grid the candidate's HJB residual is below 1.74×10^{-17}, the Fokker–Planck residual below 1.34×10^{-15}, and $|\sqrt{m}\, w - 1|$ below 1.00×10^{-15}. No proof step consumes them.

The falsifier battery. Every certificate carries 9 controls that must fire for the record to be accepted; all 9 fire on all 8 instances. Two of them attack the extension's own new lines.

control	what must happen
X1	outward rounding is load-bearing: the frozen verifier's instance-free control
X2	perturbing the candidate's mean a_0 by 10^{-2} explodes Y_0 and the radii polynomial refuses
X3	the same perturbed candidate fails the pointwise PDE witness, by more than 10^{-4} and by more than $10^6\, r$
X4	dropping a mode from the rebuilt branch variable w explodes Y_0 and the radii polynomial refuses
X5	the residual built with the parity of p forged dwarfs the true residual by 10^6 and the defect by 10^3
X6	the positivity certificate for m refuses at radius 10
X7	the positivity certificate for w refuses at radius 10 and the validation refuses for that stated reason
X8	zeroing the instance's own harmonic data term (A_2 at T1, A_3 at T6) explodes Y_0 and the radii polynomial refuses
X9	applying the data term at the wrong mode (A_2 and A_3 swapped) explodes Y_0 and the radii polynomial refuses

The peak counter carries its own red controls: a chain whose signs do not alternate, a slope sign that does not match its neighbours, a region whose sign cannot be certified, and a cell pad without the ball's derivative folded in, must each refuse. Positivity of m and of the branch selector w is certified over the whole ball, never observed on a plot. Peak counts never trust a float sign: floats propose, certified region chains decide, and anything uncertifiable refuses.

The float map, and the falsified prior. The companion atlas page draws a float phase map of 224 equilibrium solves at $N = 32$ over $(\sigma, A_2/A_1)$, pinned by sha256 (`30b3cfe49514…`); it is a guide to where the theorems sit, and no proof step consumes it. The falsified prior worth naming: the belief that equilibrium peaks cannot exceed potential wells was this program's own working assumption, formed on an unreproduced 508-sample float campaign at the origin bench—not folk belief, and not literature. The certificates above refute it.

9 Related work and fences

- Karuturi [3]—the mandatory fence: spontaneous instability of the uniform state under an interaction kernel with *no* potential, with a cognate $8\pi^2\nu^2$ constant for a different quantity.

Ours is the driven response to a structured cost under a fixed potential, with a monotone coupling.

- Cesaroni–Cirant [2]—the foil: there, density peaks mirror the potential's wells; here they certifiably exceed them.
- Osborne–Smears [4]—different in kind: a posteriori finite-element analysis, not equilibrium enclosures.
- Cecchin–Dai Pra–Fischer–Pelino [5]—non-uniqueness via a different mechanism, on a finite state space.
- van den Berg–Lessard [6] (radii polynomials) and Rump [7] (the Krawczyk operator and interval verification)—the computer-assisted-proof lineage this work instantiates.

10 Open problems

A PDE-level census (the count of Section 7 lifted from the truncation to the system); k-peak uniformity in k; sharper digits of the splitting threshold than the pin $[0.13, 0.14]$; non-even branches as $\gamma \to 0$; and the time-dependent forward–backward computer-assisted proof.

11 Status and priority

Released from the repository; not peer-reviewed; not independently rerun. Two priority claims, each stated with its fence and open to refutation:

1. To our knowledge, these are the **first certified equilibria of a mean-field game whose exact peak count strictly exceeds the potential's well count**. The nearest published phenomena are spontaneous instability [3] and peaks that mirror the potential [2]; neither carries validated-numerics certificates, and we found no validated-numerics equilibrium enclosures for mean-field games anywhere in the literature.
2. The base instance of this program—the $\sigma = 1/2$ single-harmonic enclosure of the companion page `reports/mfg-congest.html`, released together with this paper and its atlas—is, to our knowledge, the **first validated-numerics enclosure of a mean-field-game equilibrium at all**: an exact solution within an explicit radius, locally unique in the full sequence-space ball, with strictly positive density.

The companion page's embedded verifier serves as the calibration gate of every certificate here; the two are one program, released together, and the priority claims above are made by that joint release.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The records:

`certs/terra-recert-t{1..8}.json`	the 8 enclosures, dated 2026-09-01
`certs/terra-peakcount-t{1..8}.json`	the 8 certified critical-point counts
`certs/terra-sigmastar.json`	σ^*, the band-pass identity, the windows (exact rationals)
`certs/terra-bracket-table.json`	the table and the threshold pin (exact rationals)
`certs/mfg-cap-census-N{2..5}-c-12.json`	the exactly-3 census, truncation level
`certs/mfg-cap-multiplicity.json`	the disjoint uniqueness balls, dated 2026-09-02

Every certificate re-runs from the repository:

`python3 instruments/mfgcap/run_recert.py t1`	the T1 enclosure, 9 falsifiers
`node instruments/critcount/run.js t1`	exactly 2 maxima over the whole ball
`python3 instruments/mfgcap/sigmastar.py`	$\sigma^* = 1/(8\pi^2)$, exact rationals
`python3 instruments/mfgcap/bracket_table.py`	the bracket table and the threshold pin
`node labs/mfg/census.js --N 5 --c -12`	exactly 3 (6,954,073 boxes, about 587 s)
`node labs/mfg/multiplicity.js`	at least three solutions per coupling
`make test`	every battery, every red control

The macros of this document are written by `tools/paper-numbers/terra-peaks.js`, which reads the records above, rounds every bound outward, derives the digits of σ^* and r_c from the rational brackets in exact integer arithmetic, and refuses when a record no longer supports a sentence; the T1 candidate record is pinned by sha256 `88838d5a57f6…`. Repository commit `9349d8bad505`.

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